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No relative rotation detected between Corsica and Sardinia

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Forty dikes of Upper Paleozoic age have been investigated paleomagnetically on both sides of the Straits of Bonifacio to detect a possible rotation of Sardinia relative to Corsica. A primary component of magnetization has been isolated after thermal and alternating field cleaning. Eleven sites in Corsica yield a paleomagnetic mean of $D = 134.7$, $I = -11.3$ ($\alpha_{95} = 6.8$), while 11 sites from Sardinia showed a paleomagnetic mean of $D = 132.7$, $I = -1.6$ ($\alpha_{95} = 7.3$). The two directions are not significantly different at the 95% confidence level. No relative rotation between Corsica and Sardinia occurred across the Bonifacio Straits and the two islands maintained the present spatial relationship since the Upper Paleozoic. A mean paleomagnetic direction of $D = 133.7$, $I = -6.5$ ($\alpha_{95} = 5.1$) for the Sardo–Corso Massif yield a corresponding virtual mean pole position at 249.4° E, 33.7° N coincident with Upper Paleozoic data of the Mediterranean region. This data implies about 70° of counterclockwise rotation of the block respect stable Europe.

1. Introduction

Anticlockwise tectonic rotation of the islands of Corsica and Sardinia away from the southcoast of France was first suggested by Argand [1] on the basis of the fit of continental margins. Hercynian structural trends in Corsica, Sardinia and Southern France lend geological support to this concept [2].

Paleomagnetic data from Tertiary volcanics of Sardinia [3–7] confirmed this hypothesis, revealing northwesterly declinations compatible with a counterclockwise rotation during the Miocene. Upper Paleozoic paleomagnetic directions from Corsica [8,9] and Sardinia [10,11] suggest differing amounts of rotation of the two islands with respect to Europe. Corsica and Sardinia are separated by the Straits of Bonifacio (Fig. 1), and a related question is whether there has at any time been relative motion between the two islands.

Matching the western margins of the islands to the southern margin of France is straightforward for Corsica and eastern Provence, where the continental margin is steep and well defined. It is more difficult to fit Sardinia against the Gulf of Lion, because the gentle slopes of the continental margin suggest crustal extension [12], and the line of the bounding rift can only be inferred with considerable uncertainty.

Reconstructions of three types have been proposed: (1) models with Corsica and Sardinia treated as a single rigid block [2,13–16]; (2) models which invoke relative rotation between the two islands [9,12,17,18]; and (3) models which invoke fault displacement between the two islands [19–21].

The third hypothesis is difficult to reconcile with the continuity of the Corsican–Sardinian batholith across the Straits of Bonifacio [22], unless the relative displacement was very small. The fit employing a relative rotation between Corsica and Sardinia (second hypothesis) was favored by one of us [17] because it leads to a better match of the continental margins and was supported by paleomagnetic data.

We report here a new attempt to detect rotation of Sardinia relative to Corsica by paleomagnetic study of NNE–SSW trending dykes which cut the Hercynian granites of the batholith on both sides of the Straits of Bonifacio. The dykes, generally of basic lithology, are reported in literature as Permian in age on the basis of their stratigraphic position and cross-cutting relationships. Radiometric ages are not available for these dykes, but

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if the dykes are indeed contemporaneous (as suggested by geological evidence), the absolute ages are not critical to determination of the relative rotation of the two islands.

The directions of these dikes on both sides of the strait of Bonifacio were investigated by Arthaud and Matte [23] who found the same orientation, excluding any differential movement between Corsica and Sardinia.

2. Sampling

Two hundred and sixty-nine oriented hand samples were collected at 40 sites on both islands (Fig. 1); a site is generally six to eight samples from a single dyke, representing an instantaneous determination of the ancient geomagnetic field direction. By averaging paleomagnetic directions from many sites, the effects of geomagnetic secular variation may be eliminated, yielding a mean direction representative of the axial geocentric dipole field.

The sites were distributed over a relatively wide geographical area to reduce the possibility that tectonically rotated sites in a local area might bias the overall result. We report here results from 23 sites from the original collection which display a stable magnetization. Results from seventeen sites were rejected after pilot AF and thermal demagnetization revealed a complex paleomagnetic record. The rejected sites were from dykes which were more weathered and/or had a different lithology

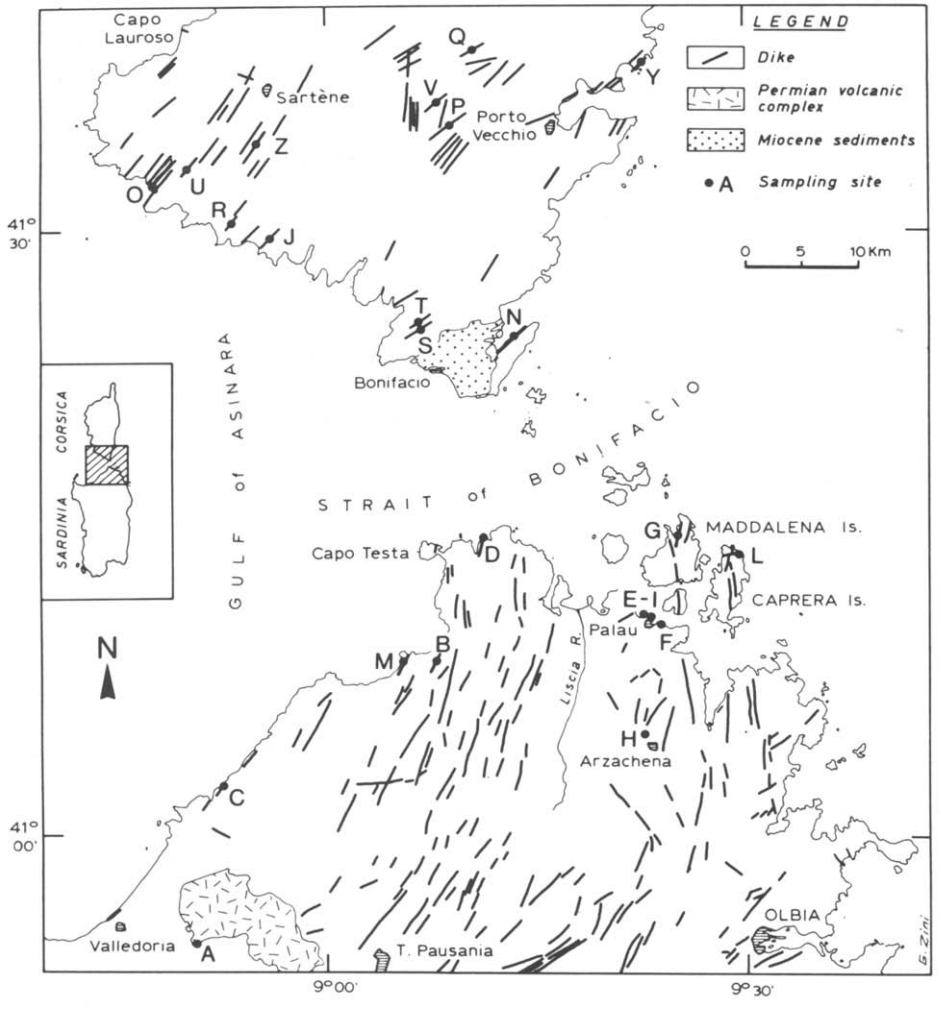


Fig. 1. Simplified geological sketch-map of the Bonifacio Straits with the sampling dikes.

than the 23 acceptable sites which form the basis of our study.

All the dikes intrude the granite of the batholith except site A which is located in the Permian volcanic complex of Gallura. No tectonic correction was employed, on the assumption that the dykes have not been disturbed from their original position. The vertical orientation of the dykes in the field support this assumption.

3. Experimental results

Measurements were made with a 2-axis SQUID magnetometer with integrated AF demagnetization coils; thermal demagnetization was done in air in a magnetically shielded furnace. One pilot specimen from each site was demagnetized in alternating fields up to 80 mT; a second pilot specimen was thermally demagnetized up to 600 °C. Representative orthogonal vector plots for AF and thermal demagnetization are shown in Fig. 2. Both AF and thermal demagnetization reveal two magnetizations: a low coercive force (H_c)/low blocking temperature (T_b) magnetization which appears to be a present field overprint, and a high H_c/T_b magnetization with a southeasterly and shallow direction (Fig. 2). We believe that this latter component is the primary magnetization. Sometimes the primary magnetization is small compared with the present field overprint, and the H_c and/or T_b spectra of these two components overlap such that the primary magnetization is not directly ob-

TABLE 1

List of Paleomagnetic data and statistical parameters of sites from Sardinia (A–M) and Corsica (N–Y).

Site	a/b	$D(°)$	$I(°)$	R	K	α_{95}
Sardinia						
A	12/18	143.5	−22.8	11.819	61	5.6
B	10/12	142.9	1.9	9.936	141	4.1
C	11/11	143.1	−9.0	10.491	20	10.6
D	9/11	134.5	−2.0	8.892	74	6.0
E	10/10	125.8	8.7	9.959	220	3.3
F	9/14	134.1	1.1	8.827	46	7.7
G	10/11	121.3	16.1	9.838	56	6.5
H	11/12	129.7	−5.1	10.759	42	7.2
I	10/10	131.6	0.7	9.888	80	5.4
L	11/13	122.1	0.8	10.780	45	6.8
M	6/11	131.6	−8.1	5.868	38	11.0
Mean	11/11	132.7	−1.6	10.754	41	7.3
Corsica						
N	11/13	148.7	−28.7	10.803	51	6.6
O	12/12	121.5	1.8	11.958	264	2.7
P	11/11	133.1	−0.9	10.823	57	6.1
Q	8/13	132.4	−2.7	7.950	139	4.7
R	7/10	136.4	−11.8	6.862	43	9.3
S	10/10	138.9	−17.7	9.909	99	4.9
T	11/11	133.2	−21.6	10.808	52	6.4
U	9/11	139.1	−4.9	8.900	80	5.8
V	11/13	136.5	−6.9	10.783	46	6.8
Z	9/9	134.7	−10.0	8.908	87	5.6
J*	5/7	158.4	−43.9	4.972	142	6.4
Y	7/10	127.9	−24.8	6.534	13	17.5
Mean	11/12	134.7	−11.3	10.781	46	6.8
Overall mean (Sardinia + Corsica excl. Site J)						
	22/23	133.7	−6.5	21.454	38	5.1

Pole position (site latitude 41.3° N; site longitude 9.1° E); Long. 249.4° E; Lat. 33.7° N; $D_p = 2.6$; $D_m = 5.1$.
a = number of specimens used in the calculation of the mean data; b = number of measured specimens.
* Excluded from the mean.

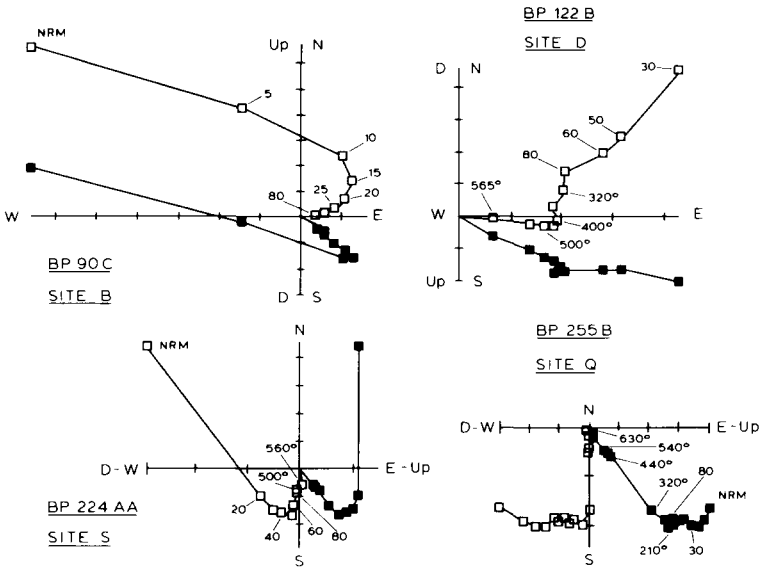


Fig. 2. Orthogonal demagnetization diagrams of two specimen from Sardinia (BP90C and BP122B) and two specimen from Corsica (BP224AA and BP255B). Open symbols for vertical projection.

served, but must be extracted using vector subtraction. To remove the viscous magnetization, we found it extremely useful to employ alternating field treatment after thermal treatment at sites C, I, M, P, Q, R, S, V, and Y.

Demagnetization data were analyzed by principal component analysis [24], with the exception of site P, whose mean was calculated using the directions after 520 ° C treatment. Mean directions of primary magnetization from 11 sites in Corsica ($D = 134.7$, $I = -11.3$; $\alpha_{95} = 6.8$) and 11 sites in Sardinia ($D = 132.7$, $I = -1.6$; $\alpha_{95} = 7.3$) are listed in Table 1 and shown in Fig. 3. Site J from Corsica was excluded from the means because its direction is anomalous with respect to the other sites for as yet unknown reasons.

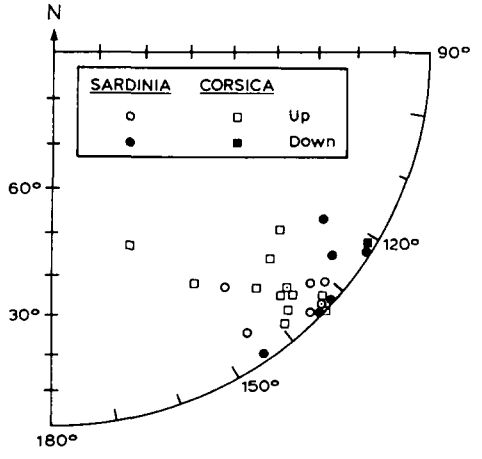


Fig. 3. Directions of magnetizations from the Upper Paleozoic dikes of Sardinia (circle) and Corsica (square). Symbols with small dot for the overall means. Open (full) symbols for negative (positive) inclinations.

Three conclusions are immediately apparent. First, the mean directions from Corsica and Sardinia are not significantly different at $P = 0.05$. Second the excellent grouping of these directions indicates that the dykes have not been significantly tilted since emplacement and magnetization. Finally, all magnetizations are reversed, consistent with a magnetization age during the Kiaman Reversed Superchron from 320–250 Ma [25].

4. Conclusions

As there is no significant difference between directions of magnetization from the northern tip of Sardinia and the southern tip of Corsica, we conclude that there has been no detectable relative rotation between Corsica and Sardinia across the Straits of Bonifacio. The kinematic movement of the two islands apparently occurred as a whole; north–south stretching that might have opened the Asinara Gulf is not recognizable by paleomagnetic data. The part of the two islands flanking the Straits of Bonifacio may be consid-

ered a single block which has maintained its present spatial relationship since the emplacement of the Kiaman-aged dykes, which on the basis of their reversed directions are consistent with a Late Paleozoic age.

A Late Paleozoic pole position at 249.4° E and 33.7° N can be calculated from the mean paleomagnetic direction of the Corso–Sardo massif ($D = 133.7$, $I = -6.5$; $\alpha_{95} = 5.1$). This pole indicates an equatorial position for Corsica and Sardinia in the Late Paleozoic, consistent with other data from the Mediterranean region. It also indicates a counterclockwise rotation of Corsica–Sardinia of about 45° with respect to the pole, and of about 70° with respect to stable Europe [26] since Paleozoic times.

Our directions agree with previously published paleomagnetic results from Permian ignimbrites in Gallura (northeast Sardinia) [10] and Permian volcanics from Nurra (Northwest Sardinia) [10,11], but they disagree with results from Permian volcanics of northern Corsica [8,9], which suggest only a small apparent rotation. Differences in declinations also with respect to southeast Sardinia (Barbagia, Ogliastra, Gerrei) [11] suggest the possibility of a decoupling between these areas during the Late Paleozoic. However, paleomagnetic results from Eocene sediments from northern Corsica [27] indicate that a larger and younger rotation of Northern Corsica has occurred. If both the Eocene and the Permian results reflect motion of the same Paleozoic granitic basement, these results indicate that the kinematic history of the Corso–Sardo microplate is more complicated than a simple counterclockwise rotation.

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